

VANCOUVER INTL, BC, Canada

WMO: 718920

Lat: 49.190N

Lon: 123.180W

Elev: 14

StdP: 14.69

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	22.7	26.6	9.0	8.7	26.7	15.7	12.2	28.9	30.3	44.1	26.2	44.7	6.7	100	0.562

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.8	77.2	65.3	74.7	64.3	72.5	63.2	66.6	74.9	65.3	73.1	63.9	71.2	8.4	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.0	86.3	71.3	61.8	82.5	69.9	60.5	78.8	68.4	31.2	75.1	30.2	73.3	29.2	71.5	74.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 24.0	20.5	17.9	DB	17.5	82.8	4.7	3.1	14.1	85.0	11.4	86.8	8.7	88.6	5.3	90.8	(4)
(5)			WB	15.5	69.3	4.9	1.8	11.9	70.6	9.1	71.6	6.3	72.7	2.7	74.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	51.1	39.9	40.6	44.3	49.0	55.4	60.3	64.8	64.8	59.3	50.7	43.9	39.1
	DBStd	10.35	6.10	5.14	4.44	3.89	4.33	3.95	3.59	3.06	3.76	4.16	5.74	5.88
	HDD50	1404	315	263	182	63	6	0	0	0	0	41	194	340
	HDD65	5171	778	683	642	480	297	150	45	41	175	442	634	804
	CDD50	1796	1	1	5	33	175	309	460	459	279	64	9	1
	CDD65	89	0	0	0	0	1	9	40	35	4	0	0	0
	CDH74	294	0	0	0	0	5	37	142	97	13	0	0	0
	CDH80	25	0	0	0	0	1	2	17	5	0	0	0	0
Wind	WSAvg	8.2	8.0	8.0	9.1	8.8	8.3	8.3	8.4	7.7	7.6	7.8	8.5	8.4
Precipitation	PrecAvg	47.6	6.3	5.0	4.4	3.0	2.6	1.9	1.5	1.6	2.5	4.8	6.9	7.3
	PrecMax	59.8	10.8	11.3	7.8	5.7	5.8	5.4	3.4	4.2	5.4	11.6	14.1	12.3
	PrecMin	33.2	1.1	1.3	1.9	0.6	0.9	0.6	0.0	0.0	0.0	0.8	1.9	2.2
	PrecStd	7.3	2.4	2.4	1.3	1.3	1.2	1.2	1.1	1.1	1.4	2.5	2.6	2.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.9	54.0	59.0	65.2	73.2	78.3	81.8	80.1	75.9	64.1	58.4	53.9
		MCWB	50.8	49.3	51.2	56.4	61.3	65.0	68.3	65.8	63.8	58.2	54.5	51.2
	2%	DB	51.4	51.3	55.1	61.1	68.5	73.9	77.6	76.6	71.3	60.8	55.4	50.4
		MCWB	49.0	47.4	49.6	53.6	59.0	63.1	66.1	65.1	62.1	55.9	52.7	48.0
	5%	DB	49.5	49.2	52.8	58.1	65.9	70.6	74.7	74.2	68.5	58.8	53.5	48.4
		MCWB	47.2	46.0	48.3	51.5	57.9	61.3	64.4	64.4	61.0	54.4	51.1	46.1
	10%	DB	47.8	47.5	50.9	55.8	63.3	68.1	72.6	72.0	66.3	57.1	51.6	46.7
		MCWB	45.8	44.7	46.9	50.0	56.3	59.7	63.2	63.5	59.6	53.4	49.4	44.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.9	50.8	52.8	57.6	62.3	66.4	69.5	67.9	65.8	59.7	55.5	51.8
		MCDB	53.0	52.6	55.9	64.0	70.8	75.6	80.6	75.7	72.3	62.5	57.0	53.5
	2%	WB	49.5	48.3	50.7	54.5	60.2	64.0	67.1	66.4	63.4	57.1	53.1	48.6
		MCDB	50.8	50.5	53.9	59.7	66.7	72.3	76.0	74.3	69.2	59.5	54.9	50.2
	5%	WB	47.5	46.5	49.0	52.5	58.6	62.2	65.4	65.3	61.8	55.2	51.5	46.5
		MCDB	49.0	48.5	52.1	57.0	64.7	69.6	73.5	72.9	67.2	57.9	53.1	48.0
	10%	WB	45.8	45.1	47.5	50.8	56.9	60.4	63.9	64.0	60.3	53.7	49.6	44.8
		MCDB	47.5	47.2	50.5	54.9	62.9	67.2	71.5	71.1	65.3	56.5	51.3	46.4
Mean Daily Temperature Range	5% DB	MDBR	8.3	9.9	10.6	11.8	12.6	12.7	13.8	13.6	12.1	9.9	9.1	8.2
		MCDBR	8.7	10.7	12.4	15.0	16.3	16.9	17.1	16.8	15.7	11.4	9.3	8.3
		MCWBR	6.8	7.6	8.5	9.9	9.8	9.4	8.9	8.6	8.8	7.6	7.4	6.7
	5% WB	MCDBR	8.1	9.3	11.6	14.0	15.5	16.4	16.3	15.7	14.5	10.1	8.6	7.9
		MCWBR	6.7	7.1	8.4	9.7	9.8	9.5	8.9	8.6	8.6	7.3	7.2	6.7
Clear-Sky Solar Irradiance	taub	0.320	0.314	0.331	0.363	0.387	0.363	0.359	0.363	0.346	0.337	0.322	0.308	
	taud	2.444	2.506	2.477	2.397	2.361	2.451	2.485	2.489	2.537	2.544	2.503	2.459	
	Ebn at Noon	233	268	281	281	277	285	284	278	272	255	232	222	
	Edh at Noon	20	24	29	34	37	34	33	31	27	23	19	17	
All-Sky Solar Radiation	RadAvg	272	545	818	1273	1665	1787	1997	1694	1174	645	321	217	
	RadStd	45	82	113	137	144	212	175	132	105	89	60	33	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (67 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page